

Rap1 Activation in Response to cAMP Occurs Downstream of Ras Activation during *Dictyostelium* Aggregation*

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We have used a doubly disrupted *rasC*[−]/*rasG*[−] strain of *Dictyostelium discoideum*, which ectopically expresses the *carA* gene, to explore the relationship between the activation of RasC and RasG, the two proteins that are necessary for optimum cAMP signaling, and the activation of Rap1, a Ras subfamily protein, that is also activated by cAMP. The ectopic expression of *carA* restored early developmental gene expression to the *rasC*[−]/*rasG*[−] strain, rendering it suitable for an analysis of cAMP signal transduction. Because there was negligible signaling through both the cAMP chemotactic pathway and the adenylyl cyclase activation pathway in the *rasC*[−]/*rasG*[−]/*[act15]:carA* strain, it is clear that RasG and RasC are the only two Ras subfamily proteins that directly control these pathways. The position of Rap1 in the signal transduction cascade was clarified by the finding that Rap1 activation was totally abolished in *rasC*[−]/*rasG*[−]/*[act15]:carA* and *rasG*[−] cells but only slightly reduced in *rasC*[−] cells. Rap1 activation, therefore, occurs downstream of the Ras proteins and predominantly, if not exclusively, downstream of RasG. The finding that *in vitro* guanylyl cyclase activation is also abolished in the *rasC*[−]/*rasG*[−]/*[act15]:carA* strain identifies RasG/RasC as the presumptive monomeric GTPases required for this activation.

The Ras subfamily proteins are monomeric GTPases that function as molecular switches by cycling between the active GTP-bound and the inactive GDP-bound state (1). In mammalian cells, activated Ras proteins impact multiple cellular signaling pathways that include mitogen-activated protein kinase cascades, phosphatidylinositol 3-kinase (PI3K)³-regulated pathways and RalGDS-dependent activation of Ral (2). These Ras-mediated responses regulate a wide range of cellular processes, including proliferation, cytoskeletal function, and differentiation. The discovery of a large number of Ras subfamily homologues in mammals and in the model organisms *Drosophila melanogaster*, *Caenorhabditis elegans*, and *Dictyostelium*

discoideum (3, 4) has raised important questions regarding the specific functions of individual Ras proteins. For example, the human Ras subfamily consists of 36 gene products that can be divided into several groups (5). The search for downstream effectors has revealed some specificity but also an enormous complexity of apparently overlapping functions (6). Thus, a major step in understanding the patterns of Ras signal transduction would be to identify the contributions made by each Ras subfamily member to a specific signaling pathway. This analysis is difficult, if there is cross-talk between the components of different pathways, or if a single input signal results in multiple responses. One useful tool in understanding Ras function is provided by examining genetically tractable organisms, such as *Dictyostelium*, that are disrupted in the *ras* gene of interest.

In the presence of nutrients, the cellular slime mold, *Dictyostelium*, grows as free-living, single-celled amoebae, but upon starvation these amoebae aggregate into a multicellular organism that progresses through a motile slug stage to form a spore mass or sorus supported by a stalk (7). The aggregation process is driven in response to cAMP, which is synthesized and secreted soon after the onset of starvation. cAMP binds to cell surface cAMP receptors (for example, the cAR1 receptor encoded by the *carA* gene), resulting in the dissociation and activation of a heterotrimeric G protein. This leads to chemotaxis to cAMP and to the activation of adenylyl cyclase A (ACA) (8).

Fourteen *Dictyostelium* Ras subfamily proteins, with a high level of amino acid identity to the mammalian H, N, and K-Ras proteins, have been described (9). The *rasC*, *rasS*, *rasG*, and *rasD* genes have all been disrupted, and each of the resulting strains has a distinct phenotype, suggesting that each encoded product performs a different function (10–14). It has been previously shown that both RasC and RasG are activated in response to cAMP, suggesting roles for both RasC and RasG in cAMP signaling (15, 16). A role for RasC and RasG has been further indicated by the finding that *rasC*[−] cells fail to aggregate and *rasG*[−] cells are delayed in aggregation (10, 14). Furthermore, ACA activation is more impacted than chemotaxis in the *rasC*[−] cells, whereas chemotaxis is more impacted than ACA activation in the *rasG*[−] cells (10). However, the loss of RasG or RasC alone did not result in a total loss of either chemotaxis or ACA activation, suggesting three possibilities: some overlap of function between RasC and RasG; a requirement of both RasC and RasG for optimal chemotaxis and optimal ACA activation; or the existence of an additional pathway for cAMP signaling. The recent finding that Rap1 is activated in response to cAMP (17) provides a potential mechanism for the third possibility:

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³ The abbreviations used are: PI3K, phosphatidylinositol 3-kinase; ACA, adenylyl cyclase A; PKB, protein kinase B; GTPγS, guanosine 5'-3-O-(thio)triphosphate; RBD, Ras-binding domain; GST, glutathione S-transferase.

signaling through Rap1 acting in parallel to signaling through RasC or RasG. To eliminate this possibility, we have ectopically expressed the *carA* gene in a doubly disrupted *rasC⁻/rasG⁻* strain. This ectopic expression was necessary, because the *rasC⁻/rasG⁻* strain exhibited greatly reduced expression of *gpaB* and *carA*, the genes encoding the $G\alpha_2$ subunit of the heterotrimeric G protein and the cAMP receptor cAR1, respectively, and was deficient in the developmentally induced production of ACA (10). The absence of these essential components for cAMP signaling rendered the *rasC⁻/rasG⁻* strain inappropriate for an analysis of signal transduction. We report experiments that show that early developmental gene expression is restored in the *rasC⁻/rasG⁻/[act15]:carA* mutant, but these cells exhibit negligible chemotaxis and cAMP relay, conclusively demonstrating the essential role of RasC or RasG in these processes. In addition, an examination of cAMP-dependent Rap1 activation in the various *ras* null strains provided evidence that Rap1 activation occurs downstream of RasG. Finally, the availability of the *rasC⁻/rasG⁻/[act15]:carA* strain enabled us to demonstrate the essential role of RasC and RasG in guanylyl cyclase activation.

EXPERIMENTAL PROCEDURES

Dictyostelium Strains—Generation of *rasG⁻*, *rasC⁻*, and *rasC⁻/rasG⁻* strains, all in the JH10 background, has been described previously (10). To generate *rasC⁻/rasG⁻/[act15]:carA* strain, 20 μ g of the PcAR1-B18 expression vector, which contains *carA* cDNA under the control of the constitutively active *actin 15* promoter and a neomycin-resistance cassette (18), was electroporated into *rasC⁻/rasG⁻* cells as previously described (19). Clones containing the selectable marker were isolated 8–15 days following the application of the selection conditions and maintained in HL5 media supplemented with 10 μ g/ml G418 (Invitrogen).

Cell Culture and Development—The JH10 strain was grown at 22 °C in HL5 medium (20) containing 50 μ g/ml streptomycin sulfate (Sigma) and 100 μ g/ml thymidine (Sigma). The various *ras* gene-disrupted strains were grown in the same medium without thymidine. In addition, the *rasC⁻/rasG⁻* strain was grown in the presence of 10 μ g/ml blasticidin S (Calbiochem), and the *rasC⁻/rasG⁻/[act15]:carA* strain was grown in the presence of 10 μ g/ml blasticidin S and 10 μ g/ml G418 (Invitrogen). Because the *rasG⁻*, *rasC⁻/rasG⁻*, and *rasC⁻/rasG⁻/[act15]:carA* cells grow poorly in shaken suspension, all strains were grown in Nunc tissue-culture plates.

To obtain cAMP-pulsed cells, wild-type and mutant vegetative cells were harvested, washed twice in KK2 (20 mM potassium phosphate, pH 6.1), and resuspended in KK2 to a final density of 5×10^6 cells/ml. 30 ml of this cell suspension was shaken at 150 rpm for 1 h and then pulsed with 100 μ l of cAMP to a final concentration of 50 nM every 6 min for 5 h. To observe aggregation streaming, vegetative or 5-h cAMP-pulsed cells were washed twice in Bonner's salts (10 mM NaCl, 10 mM KCl, 2 mM CaCl_2), seeded at $\sim 4 \times 10^5$ cells/cm² in Nunc tissue culture dishes submerged under Bonner's salts, incubated at 22 °C, and photographed with a Leica DMIL inverted microscope (Leica Microsystems, Wetzlar, Germany), a Leica DFC350F camera

(Leica Microsystems), and Openlab software (Improvision, Coventry, England).

cAMP Binding Assay—cAMP-pulsed cells were treated with 2 mM caffeine for 30 min, washed twice in ice-cold KK2, and resuspended to a density of 1.5×10^8 cells/ml. The binding of [³H]cAMP to the cell surface was measured by the ammonium sulfate stabilization assay (21). Cells (1.5×10^7) were incubated with 100 μ l of KK2 containing 10 nM [³H]cAMP (0.9 TBq/mmol, GE Healthcare, Little Chalfont, Buckinghamshire, England) and 5 mM dithiothreitol (Sigma) for 1 min at 0 °C. 1 ml of saturated ammonium sulfate was then added, followed immediately by the addition of 100 μ l of bovine serum albumin (10 mg/ml, Sigma). After 5-min incubation at 0 °C, the samples were centrifuged at $8000 \times g$ for 2 min. The supernatant was removed, and the pellet was resuspended in 110 μ l of 1 M acetic acid. 100 μ l of the resuspended pellet was mixed with 4 ml of scintillation mixture (Ready Safe liquid scintillation cocktails, Beckman Coulter, Fullerton, CA), and the amount of bound [³H]cAMP was measured using a Beckman LS6000IC scintillation counter.

Chemotaxis Assays—For cAMP micropipet assays, cells were pulsed with cAMP as described above, washed twice with Bonner's salts, and then seeded in Nunc tissue culture dishes at a cell density of $\sim 5 \times 10^5$ cell/cm² in Bonner's salts. At $t = 0$, an Eppendorf Femtotip Micropipet (Hamburg, Germany) filled with 100 μ M cAMP was placed in the field of view, and cell movements toward the micropipet were monitored by time-lapse video microscopy using an Olympus IX-70 inverted microscope, a DAGE-MTI CCD-100 camera (Michigan City, IN), and Scion Image 4.0 software (Frederick, MD). Instantaneous velocities and chemotaxis indices were determined as described previously (22).

Western Blot Analyses and PKB Phosphorylation Assays—Cells were pelleted by centrifugation and lysed in 1 $\times$ Laemmli SDS-PAGE loading buffer (6 $\times$ buffer: 350 mM Tris-Cl, pH 6.8, 10% SDS, 600 mM dithiothreitol, 0.012% w/v bromphenol blue, 30% glycerol) by boiling for 5 min (23). 10 μ g of protein was then fractionated by SDS-PAGE. After electrophoresis, the proteins were transferred electrophoretically onto Hybond-P polyvinylidene difluoride membranes (Amersham Biosciences), which were then blocked with nonfat milk (23) and probed with the appropriate antibody. Antibody binding was detected by enhanced chemiluminescence (ECL, GE Healthcare). PKB phosphorylation was determined as described previously (14). Briefly, cAMP-pulsed cells were washed twice in KK2, resuspended to 5×10^7 cells/ml in KK2, and then stimulated by addition of cAMP to a final concentration of 100 nM. Aliquots of 100 μ l were removed at intervals, before and after stimulation, and mixed with 20 μ l of 6 $\times$ SDS-PAGE loading buffer. 10 μ g of protein was fractionated by SDS-PAGE and then subjected to Western blot analysis using a phosphothreonine-specific antibody (Cat#9381, Cell Signaling Technologies, Danvers, MA). Equal sample loading was verified by staining a duplicate gel with Coomassie Blue. To assess equal PKB expression levels in all strains, Western blots were also analyzed using a PKB-specific antibody. The protein concentration was determined using the protein assay from Bio-Rad.

Role of Ras in Chemotaxis and cAMP Relay

Northern Blot Analyses—For Northern blot analyses, cells were pulsed with cAMP as described above, and total RNA was extracted at intervals using guanidinium isothiocyanate (24). 15- μ g aliquots were size-fractionated on 1.25% agarose-formaldehyde gels, blotted onto Hybond-N⁺ membrane (Amersham Biosciences), and probed with DNA fragments from plasmids carrying the specific genes of interest, as previously described (25).

Adenylyl Cyclase Assay—*In vitro* adenylyl cyclase activity was determined as described previously (14), except that cell suspensions were frozen on dry ice and thawed, and then vortexed in the presence of glass beads (<106 μ m, Sigma). cAMP was recovered by sequential chromatography through Dowex and Alumina columns, and the eluted [³²P]cAMP was measured using a Beckman LS6000IC scintillation counter.

cGMP Production and Guanylyl Cyclase Activation—To measure cGMP production, cells were pulsed with cAMP as described above, washed twice with KK2, and resuspended to a density of 1×10^8 cells/ml in KK2 containing 2 mM caffeine. The cells were stimulated with 100 nM cAMP, and 100- μ l samples were lysed at intervals by the addition of 100 μ l of 3.5% perchloric acid, followed by the addition of 50 μ l of 50% saturated KHCO₃. cGMP levels were then measured using a [³H]cGMP assay kit (GE Healthcare, TRK500).

In vitro guanylyl cyclase assay was determined as described previously (26). Briefly, cAMP-pulsed cells were washed and resuspended to a density of 2×10^8 cells/ml in KK2. One volume of cell suspension was mixed with one volume of 10 mM Tris-Cl, pH 8, 1.5 mM EGTA, 500 mM sucrose, in the presence or absence of 200 μ M GTP γ S, and immediately lysed by pushing the cells through a 5 μ M Isopore Membrane Filter (Millipore, Ireland) placed directly between a 1-ml syringe and a B-D 26G1/2 needle (BD Biosciences). 150 μ l of cell lysate was mixed with an equal volume of assay mix (10 mM Tris-Cl, pH 8, 250 mM sucrose, 40 mM dithiothreitol, 1 mM GTP, 5 mM MgCl₂), and the mixture was incubated at 22 °C for 60 s. Aliquots (100 μ l) were mixed with 100 μ l of 3.5% perchloric acid, followed by addition of 50 μ l of 50% saturated KHCO₃. cGMP levels were then measured using a [³H]cGMP assay kit (GE Healthcare TRK500).

RasG, RasC, and Rap1 Activation Assays—The Ras-binding domain (RBD) of Byr2 was expressed in *Escherichia coli* as a GST fusion protein and purified as described previously (15). The purified GST-Byr2-RBD was used for the detection of activated proteins. After cAMP pulsing, cells were harvested by centrifugation and resuspended at a density of 2×10^7 cells/ml in KK2 containing 1 mM caffeine. After 30 min, aliquots (2 ml) of cell suspension were stimulated by addition of cAMP to a final concentration of 15 μ M. Cell suspensions (350 μ l) were lysed at the indicated times by mixing with an equal volume of $2 \times$ lysis buffer (20 mM sodium phosphate, pH 7.2, 2% Triton X-100, 20% glycerol, 300 mM NaCl, 20 mM MgCl₂, 2 mM EDTA, 2 mM Na₃VO₄, 10 mM NaF, containing two tablets of protease inhibitor mixture (Roche Diagnostics) per 50-ml buffer). The lysates were centrifuged for 10 min, and the protein concentrations of the supernatants were determined using the protein assay from Bio-Rad. 400 μ g of protein was incubated with 100 μ g of GST-Byr2-RBD on glutathione-Sepharose beads (GE Healthcare Bio-sciences AB, Uppsala, Sweden) at 4 °C for 1 h.

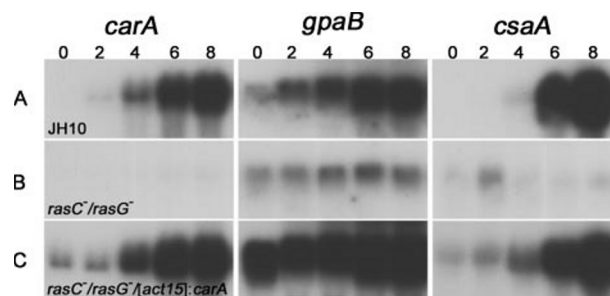


FIGURE 1. Northern blot analysis of early developmental gene expression. RNA was isolated from JH10 (A), *rasC*[−]/*rasG*[−] (B), and *rasC*[−]/*rasG*[−]/*[act15]:carA* (C) cells that had been pulsed with cAMP for the indicated times (in hours), and then subjected to electrophoretic fractionation. Blots were hybridized with cDNA probes corresponding to the indicated genes.

TABLE 1

Cell surface cAMP binding

Strain	cAMP bound ^a
	nmol/10 ⁷ cells
JH10	4.3 ± 0.4
<i>rasC</i> [−] / <i>rasG</i> [−]	0.8 ± 0.2
<i>rasC</i> [−] / <i>rasG</i> [−] / <i>[act15]:carA</i>	6.6 ± 0.5

^a The values are the means of two experiments.

The beads were harvested by centrifugation and washed three times in $1 \times$ lysis buffer. 50 μ l of $1 \times$ SDS gel loading buffer was then added to the pelleted beads, and the suspension was boiled for 5 min. Samples were subjected to SDS-PAGE and Western blot analysis with anti-RasG, RasC, or Rap1 antibodies.

RESULTS

Restoration of Early Developmental Gene Expression in *rasC*[−]/*rasG*[−] Strain by the Ectopic Expression of *carA*—Characterization of a *rasC*[−]/*rasG*[−] double mutant by Northern blot analysis had revealed that expression of the early developmentally genes, *carA*, *gpaB*, and *csaA*, was dramatically reduced, with the effect on *carA* expression being the most pronounced (10). Thus, any signaling phenotype observed in this strain could be due to insufficient levels of essential developmentally expressed signaling proteins, such as cAR1, G α _s, and ACA. Because *carA* expression was the most defective, we expressed the gene encoding cAR1 (*carA*) from the constitutively active *act15* promoter in the *rasC*[−]/*rasG*[−] strain, in an attempt to provide sufficient cAR1 to allow at least some cAMP signaling. As anticipated, *carA* was now clearly expressed in vegetative cells of this strain, whereas, in contrast it was undetectable in vegetative JH10 and *rasC*[−]/*rasG*[−] cells. Surprisingly *gpaB* and *csaA* were also expressed in vegetative *rasC*[−]/*rasG*[−]/*[act15]:carA* cells at much greater levels than in JH10 or *rasC*[−]/*rasG*[−] cells. Even more surprisingly, the developmental expression of *carA*, *gpaB*, and *csaA* was restored by the ectopic expression of *carA* (Fig. 1, A and C). Thus, the presence of cAR1 in vegetative cells at the onset of development is a positive regulator of early gene expression and is able to offset the absence of RasG and RasC, normally required for this induction. Because early gene expression was restored in the *rasC*[−]/*rasG*[−]/*[act15]:carA* strain, it enabled an investigation of the possible roles of other Ras proteins in cAMP-induced signal transduction.

To demonstrate that the restored *carA* expression in the *rasC*[−]/*rasG*[−]/*[act15]:carA* strain produced functional cAR1, we determined the binding of cAMP to the cell surface of

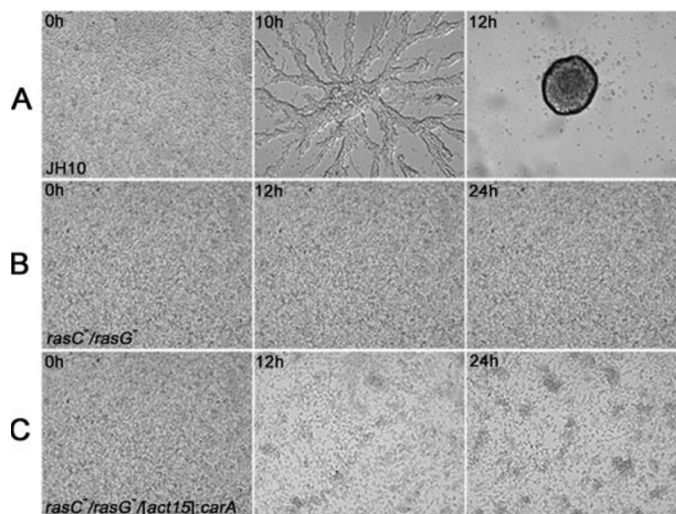


FIGURE 2. **Aggregation on a plastic surface of cells submerged in Bonner's salts solution.** Strains JH10 (A), *rasC*[−]/*rasG*[−] (B), and *rasC*[−]/*rasG*[−]/*act15::carA* (C) were photographed at the indicated times after plating. Single experiments are shown, but the results for the strains were highly reproducible.

cAMP-pulsed cells. The results, shown in Table 1, indicated that cAMP binding was reduced, as expected, in the *rasC*[−]/*rasG*[−] relative to JH10, but restored to greater than wild-type levels in the *rasC*[−]/*rasG*[−]/*act15::carA* cells. This result demonstrated that the overexpression of *carA* indicated in Fig. 1 resulted in functional protein being localized on the surface of the *rasC*[−]/*rasG*[−]/*act15::carA* cells.

As was the case with the original *rasG*[−] cells (12), the new *rasG*[−] strains grew poorly in shaken suspension and were multinucleate (data not shown). The *rasC*[−]/*rasG*[−] and *rasC*[−]/*rasG*[−]/*act15::carA* cells exhibited no growth whatsoever in shaken suspension and grew less efficiently than *rasG*[−] cells on a plastic surfaces (data not shown). These results indicate that RasC is capable of performing some of the RasG functions in vegetative cells. In addition, it was shown previously that, although *rasC*[−] cell grew normally, they exhibited deficiencies in some vegetative cell functions, indicating an important role for RasC in vegetative cells (27).

Developmental Phenotype—When grown in association with bacteria, the *rasC*[−]/*rasG*[−]/*act15::carA* cells did not form aggregates on bacterial lawns (data not shown). In addition, aggregation streams were not observed when cells were plated in plastic dishes under non-nutrient buffer, although small cell clumps were formed after prolonged incubation (Fig. 2C), results similar to those observed previously for AX2/*rasC*[−] (14) and JH10/*rasC*[−] cells (10). The *rasC*[−]/*rasG*[−] cells did not form small clumps under these conditions (Fig. 2B). The formation of aggregation streams by the control JH10 cells was observed 10 h after plating, and typical tight aggregates formed by 12 h (Fig. 2A). When pulsed with cAMP for 5 h before plating in plastic dishes, *rasC*[−]/*rasG*[−]/*act15::carA* exhibited the same phenotype as the non-pulsed cells (data not shown). Thus, cells containing neither RasG nor RasC are incapable of aggregation under any of the experimental conditions tested, and this defect is not due to the inability to express the early developmental

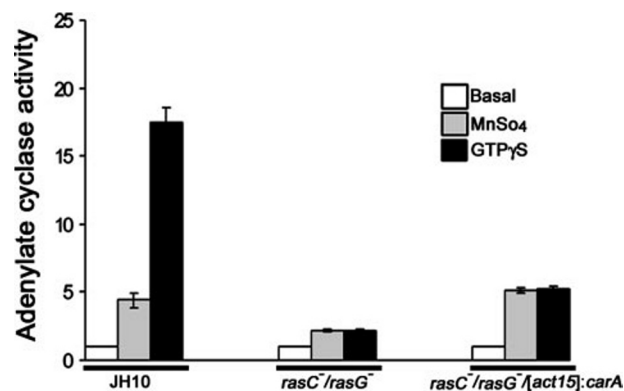


FIGURE 3. **In vitro GTPγS stimulation of adenyl cyclase activity.** JH10, *rasC*[−]/*rasG*[−], and *rasC*[−]/*rasG*[−]/*act15::carA* cells were pulsed with cAMP for 5 h, and lysates were assayed for ACA activity in the presence of 5 mM MnSO₄ (gray bar), 40 μM GTPγS (black bar), or no additional component (white bar). Plotted values are normalized relative to the unstimulated activity in the absence of MnSO₄ or GTPγS (white bar). The means and standard deviations for three independent experiments are shown.

genes (Fig. 1C). These results show that aggregation is totally blocked in the absence of both RasC and RasG. To determine the role played by these proteins in the cAMP relay and chemotaxis signal pathways, these downstream responses to cAMP were further explored.

The Presence of Either RasG or RasC Is Essential for ACA Activation—The activation of ACA by signaling through the heterotrimeric G protein is essential during aggregation to provide the extracellular cAMP to relay the signal through the population (8). GTPγS activates ACA by uncoupling the Gα subunits from the Gβγ subunits without the requirement for receptor occupation by cAMP (8, 28). ACA activity in the absence of GTPγS was reduced in the lysates of cAMP-pulsed *rasC*[−]/*rasG*[−] cells (Fig. 3), suggesting that these cells expressed less ACA, a result consistent with the defect in early gene expression (Fig. 1). In contrast, lysates prepared from cAMP-pulsed *rasC*[−]/*rasG*[−]/*act15::carA* cells, had wild-type levels of ACA activity. When stimulated with GTPγS, there was no increase in ACA activity in either *rasC*[−]/*rasG*[−] or in *rasC*[−]/*rasG*[−]/*act15::carA* cells (Fig. 3). This result indicates that *rasC*[−]/*rasG*[−]/*act15::carA* cells are incapable of cAMP-induced signal relay and is in marked contrast to those observed previously for cells with single *ras* gene disruptions, which exhibit some ACA activation (10, 14). Thus, signal transduction from the heterotrimeric G protein to ACA must involve either RasC or RasG.

The Presence of Either RasG or RasC Is Essential for Chemotaxis, PKB Phosphorylation, and Guanylyl Cyclase Activation—To investigate the requirement of RasC and RasG for chemotaxis, cells that had been pulsed for 5 h with cAMP were subjected to a spatial gradient created by diffusion of cAMP from the tip of a micropipet. Under these conditions, chemotaxis-competent cells respond by migrating toward the tip (29). Chemotactic indices for the *rasC*[−]/*rasG*[−] and the *rasC*[−]/*rasG*[−]/*act15::carA* cells revealed the virtual absence of chemotaxis in these two strains (Table 2). Thus, cAMP-induced chemotaxis toward cAMP requires signal transduction through either RasC or RasG. The results for the double *ras* gene deletion strains are in marked contrast to

TABLE 2

Chemotaxis analysis of *Dictyostelium* cells in a spatial gradient of cAMP

All values are means $\pm$ S.D. for the indicated number of cells.

Strain	Number of cells	Instantaneous velocity $\mu\text{m}/\text{min}$	Chemotaxis index ^a
JH10	20	11.18 $\pm$ 1.51	0.62 $\pm$ 0.12
<i>rasG</i> [−]	21	5.98 $\pm$ 1.73	0.45 $\pm$ 0.14
<i>rasC</i> [−]	22	12.97 $\pm$ 3.99	0.71 $\pm$ 0.15
<i>rasC</i> [−] / <i>rasG</i> [−]	20	6.60 $\pm$ 3.33	0.05 $\pm$ 0.02
<i>rasC</i> [−] / <i>rasG</i> [−] / <i>act15::carA</i>	20	5.91 $\pm$ 3.41	0.04 $\pm$ 0.01

^a Chemotaxis index was calculated as the net distance traveled toward the source of chemoattractant divided by the total distance traveled in that time period.

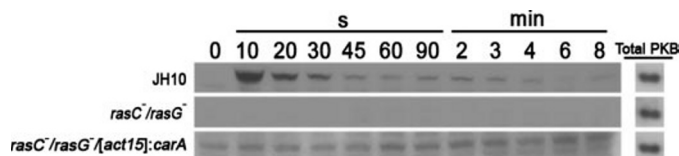


FIGURE 4. PKB phosphorylation in response to cAMP. Extracts from JH10, *rasC*[−]/*rasG*[−], and *rasC*[−]/*rasG*[−]/*act15::carA* cells that had been pulsed with cAMP for 5 h were prepared at the indicated times after a single 100 nM cAMP stimulus and subjected to Western blot analysis using a phosphothreonine-specific antibody.

those obtained previously for the single *ras* gene deletion strains, which both exhibited appreciable chemotaxis (see Table 2 for comparative values). The calculated average instantaneous velocities of the *rasC*[−]/*rasG*[−] and *rasC*[−]/*rasG*[−]/*act15::carA* cells were lower than those for the wild type, although the values clearly indicate that both strains are still capable of random motility (Table 2).

It has been suggested that PI3K-mediated signaling pathway is important for *Dictyostelium* chemotaxis (30–34). cAMP stimulation of aggregation competent cells leads to an increase in phosphatidylinositol 3,4,5-triphosphate at the leading edge of chemotaxing cells followed by translocation and activation of PH domain containing protein such as PKB to the site of PI3K activity (32, 35, 36). To test the role of RasC and RasG in PI3K activation we monitored cAMP-induced PKB phosphorylation, because PKB is one of the downstream targets of PI3K. Lysates of cAMP-stimulated *rasC*[−]/*rasG*[−]/*act15::carA* cells were probed by Western blotting using a phosphothreonine-specific antibody, which is capable of unambiguously detecting the phosphorylation of PKB (14). PKB was transiently phosphorylated after 10 s in response to cAMP in JH10 cells, and the phosphorylation level then decreased, with a second, smaller increase occurring after 120 s (Fig. 4). In *rasC*[−]/*rasG*[−] and *rasC*[−]/*rasG*[−]/*act15::carA* cells, the transient activation of PKB was totally eliminated, although the basal level of phosphorylated PKB was higher in the latter strain. The total amounts of PKB protein, as detected using a specific PKB antibody, were identical in all three strains. These results demonstrated that all cAMP-dependent signal transduction leading to PKB phosphorylation occurs through either RasG or RasC.

In *Dictyostelium*, cGMP production has also been shown to be associated with the chemotactic response (37, 38). We therefore examined cGMP production in response to cAMP in the *ras* mutant strains. In the JH10 parental cell line, there was a burst of cGMP production 10 s after the application of

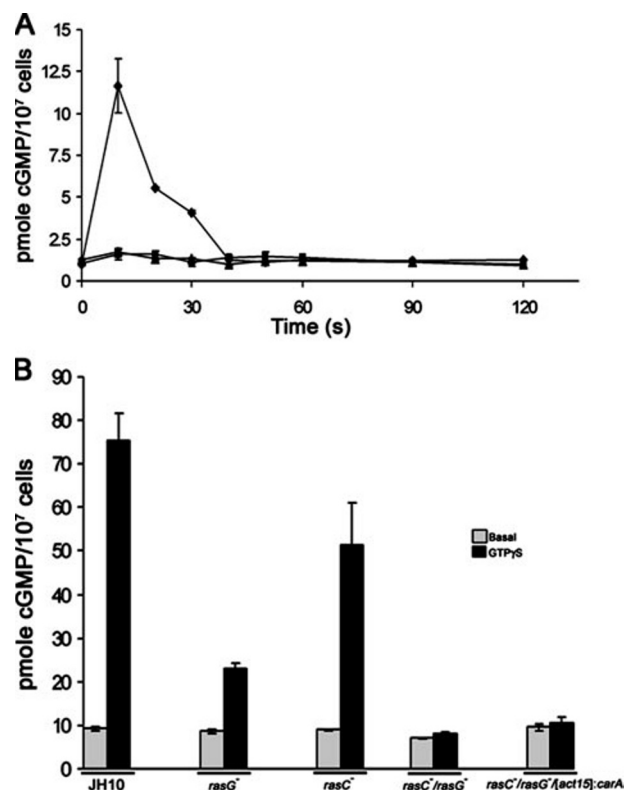


FIGURE 5. cGMP production in response to cAMP and *in vitro* GTP γ S stimulation of guanylyl cyclase activity. A, JH10 ($\diamond$), *rasC*[−]/*rasG*[−] ($\blacksquare$), and *rasC*[−]/*rasG*[−]/*act15::carA* ($\blacktriangle$) cells were pulsed for 5 h with cAMP, washed, and then stimulated with 100 nM cAMP. Samples were taken at the indicated times and assayed for cGMP accumulation (see “Experimental Procedures”). The means and standard deviations for three independent experiments are shown. B, JH10, *rasG*[−], *rasC*[−], *rasC*[−]/*rasG*[−], and *rasC*[−]/*rasG*[−]/*act15::carA* cells were pulsed with cAMP for 5 h, and lysates were then assayed for guanylyl cyclase activity in the absence (gray bar) or the presence of GTP γ S (black bar). The means and standard deviations for three independent experiments are shown.

cAMP, which then returned to the basal level after $\sim$ 30 s. However, this response was not detectable in *rasC*[−]/*rasG*[−]/*act15::carA* cells or *rasC*[−]/*rasG*[−] cells (Fig. 5A), consistent with a requirement of either RasC or RasG for the activation of guanylyl cyclases. It had been shown previously that there was still appreciable cGMP production in response to cAMP in the single *ras* gene deletion strains (10).

GTP γ S stimulates *in vitro* guanylyl cyclase activity (39), and an involvement of a monomeric GTPase has been proposed (40). We therefore investigated the possibility that either RasC or RasG might be this presumptive protein, by examining the GTP γ S stimulation of guanylyl cyclase activity in the various *ras* mutant strains. GTP γ S stimulated the guanylyl cyclase activity of JH10 lysates $\sim$ 8-fold, but stimulation was not detectable in either the *rasC*[−]/*rasG*[−] and *rasC*[−]/*rasG*[−]/*act15::carA* cell lysates (Fig. 5B). These results are consistent with the conclusion that all cAMP signal transduction responsible for guanylyl cyclase activation is mediated through pathways involving either RasG or RasC. The guanylyl cyclase activity of *rasC*[−] lysates was stimulated $\sim$ 5.5-fold and that of *rasG*[−] lysates $\sim$ 2.5-fold (Fig. 5B), consistent with the idea that RasG is more important than RasC for guanylyl cyclase activation (10).

Rap1 Activation—To understand the relationship between Rap1 activation on the one hand and RasG and RasC activation

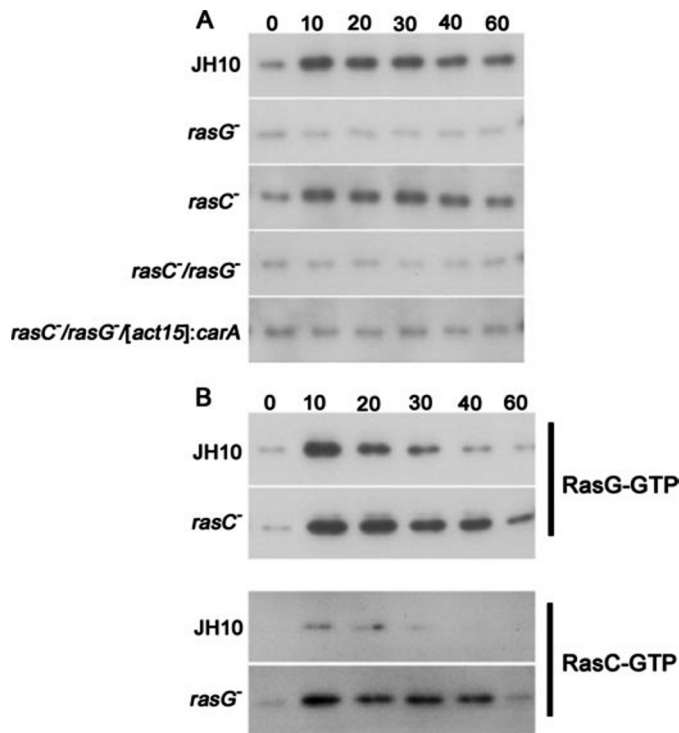


FIGURE 6. Ras subfamily protein activation in response to cAMP. A, extracts from JH10, *rasG*⁻, *rasC*⁻, *rasC*⁻/*rasG*⁻, and *rasC*⁻/*rasG*⁻/*act15*:*carA* cells, that had been pulsed with cAMP and then subjected to a single cAMP stimulus for the indicated times (in seconds), were bound to GST-Byr2-RBD. The bound material was analyzed by Western blots, using an antibody specific for Rap1. B, extracts from JH10, *rasG*⁻, and *rasC*⁻ cells, that had been pulsed with cAMP and then subjected to a single cAMP stimulus for the indicated times (in seconds), were bound to GST-Byr2-RBD. The bound material was analyzed by Western blots, using antibodies specific for RasG and RasC.

on the other, Rap1 activation was determined in the various *ras* null strains. Strains were pulsed with cAMP for 5 h and then stimulated with cAMP. Cells were lysed, and the level of GTP-bound Rap1 was determined by an affinity “pull-down” assay (15). There was a high basal level of activated Rap1 in JH10 cells, but there was an appreciable further activation in response to cAMP (Fig. 6A). Rap1 activation in response to cAMP was totally eliminated in *rasG*⁻ cells, in *rasC*⁻/*rasG*⁻, and in *rasC*⁻/*rasG*⁻/*act15*:*carA* cells. In contrast, Rap1 activation was only slightly reduced in *rasC*⁻ cells compared with JH10 cells (Fig. 6A). These data suggest that signaling through RasG is essential for Rap1 activation, although signaling through RasC may play a part.

RasG and RasC Activation—To investigate possible cross-talk between the RasG- and RasC-mediated signal transduction pathways, we determined whether either RasC or RasG was differentially activated in the *rasG*⁻ and *rasC*⁻ mutant strains. RasG was rapidly activated in *rasC*⁻ cells with a peak of activation at ~10 s, which was similar to that for JH10 cells, suggesting no compensation. However, the subsequent loss of activated RasG was less pronounced (Fig. 6B) suggesting an alteration in the adaptation component of the signaling process. In *rasG*⁻ cells, the activation of RasC was enhanced (Fig. 6B), suggesting a compensatory role for RasC in these cells.

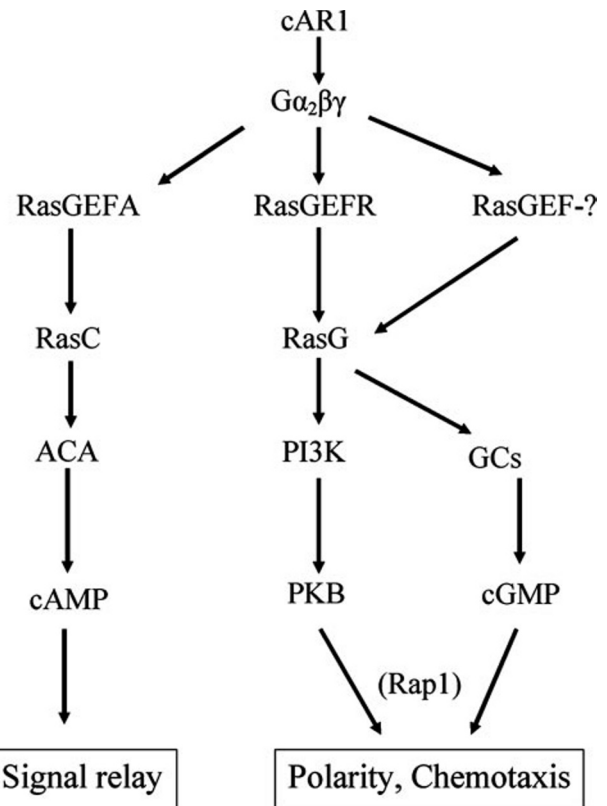


FIGURE 7. Model of cAMP signaling via RasC and RasG. The cAMP-signaling pathway branches at the level of RasGEF specificity. Signaling through guanylyl cyclase (GC) and signaling through PI3K appear to constitute parallel pathways during the chemotactic response to cAMP. Rap1 is activated downstream of RasG, but its precise site of action is as yet unknown.

DISCUSSION

During *Dictyostelium* aggregation, cAMP induces both the activation of ACA, and chemotaxis (Fig. 7). Both processes require the cAMP receptor (cAR1) and the associated heterotrimeric G protein (Gα₂βγ) (41), and we have previously shown that RasC and RasG act downstream of Gα₂βγ and upstream of the other known components (10, 14, 15). The available evidence suggests that signaling through RasC is more important for the activation of ACA, whereas signaling through RasG is more important for chemotaxis (10). However, neither ACA activation nor chemotaxis was completely eliminated in either the *rasC*⁻ or the *rasG*⁻ strains. Because the defects in the single null strains were not simply additive, we favored the hypothesis that there was some overlap of function (10). However, there are at least two other possibilities: that both RasC and RasG are required for both optimal chemotaxis and optimal ACA activation, or that an additional Ras protein, such as Rap1, recently shown to be activated in response to cAMP (17), is also important at this point in the signal transduction pathway.

The finding that early gene expression is markedly repressed in the *rasC*⁻/*rasG*⁻ strain indicates an important role for RasC and RasG in response to starvation that induces early gene expression. Expression of *carA* gene from the constitutive *actin 15* promoter markedly elevates the levels of *carA* expression in vegetative cells. This observed expression clearly restores the developmentally regulated expression of *carA* itself and that of

other developmental genes, that was suppressed in the *rasC⁻/rasG⁻* strain. The mechanism by which early gene expression is repressed in the *rasC⁻/rasG⁻* strain and the mechanism by which such expression is restored in the *rasC⁻/rasG⁻/[act15]:carA* strain were not pursued in this study, which was designed specifically to look at the role of Ras proteins in the signal transduction pathway.

The experiments using the *rasC⁻/rasG⁻/[act15]:carA* mutant strain show conclusively that all chemotactic and signal relay responses to cAMP are mediated through either RasG or RasC, and the results negate the need for an additional Ras protein at this step of the pathway (Fig. 7). The partial chemotaxis and ACA activation observed in the single *ras* null mutants could be due to an overlap of RasC and RasG function, because there is close amino acid similarity between the two proteins. Thus, there could be an interaction of the one Ras protein with the effectors of the other. In the wild-type strain, each Ras protein would out-compete the other for its own effectors. However, in the single null strains, where one protein is missing, the competition would be removed, allowing an interaction of the remaining Ras protein with both sets of effectors. Consistent with this idea of a compensatory mechanism, is the finding that there is an increase in RasC activation in *rasG⁻* cells (Fig. 6B), although there was no reciprocal increase in RasG activation in *rasC⁻* cells (Fig. 6B).

It is also possible that there is a dual role for RasG and RasC in both ACA activation and in chemotaxis. This idea is attractive, because a requirement for the two Ras proteins for ACA activation could explain the enhancement in the cAMP signal relay as aggregation progresses. Thus, RasG might be responsible for initiating the process, but RasC might be important for the amplification of the signal that is needed to entrain the cells into aggregation streams. Similarly, if there is a dual role for RasG and RasC in chemotaxis, RasC might be required early in aggregation for a response to low concentrations of cAMP and RasG required later in aggregation for a response to higher concentrations of cAMP. This interpretation is consistent with the delayed aggregation by the *rasG⁻* cells and the absence of aggregation of the *rasC⁻* cells. However, because there is as yet no experimental support for these possibilities, they have not been included in our model (Fig. 7).

Rap1 overexpression leads to shape changes and enhanced cell adhesion of vegetative *Dictyostelium* cells (42, 43). In addition, the available evidence suggests that Rap1 is essential for vegetative cell viability (44). Activated Rap1 is capable of binding to the RBD of Phg2, and it has been suggested that this binding leads to an increase in cell adhesion (45) and plays a role in cell motility (17). It has also been shown that Rap1 is activated by the RasGEF activity of GbpD in vegetative cells (45). A potential role for Rap1 during aggregation has been recently suggested by the finding that the protein is activated in response to cAMP in aggregation competent cells (17). This activation was absent in *rasC⁻/rasG⁻/[act15]:carA* and *rasG⁻* cells, indicating that it occurs downstream of signaling through RasG. Because we have no evidence for the exact role of Rap1 in this signal transduction

pathway, we have simply placed it downstream of RasG and upstream of polarity/chemotaxis (Fig. 7). Rap1 activation in response to cAMP was slightly reduced in *rasC⁻* cells, and we therefore can't rule out the possibility that some signaling is mediated through RasC, but this has also not been considered in our model (Fig. 7). The levels of activated Rap1 that were present prior to the cAMP stimulation were also reduced in *rasG⁻* cells but not reduced in *rasC⁻* cells, indicating a role for RasG in regulating basal levels of Rap1 activation. Because GbpD is involved in Rap activation in vegetative cells (45), it might also be involved in the activation of Rap1 in chemotaxing cells in response to cAMP. It will be interesting to investigate the nature of the link between RasG and GbpD.

Because RasG interacts with the RBD of two of the *Dictyostelium* PI3Ks (PI3K1 and PI3K2), it was postulated that RasG is a regulator of PI3K activity (46). Initially it was suggested that the production of PIP3 by the activity of PI3K plays a pivotal role in the chemotactic response to cAMP (35, 36, 47). However, recently there have been two reports indicating that strains with several deleted *pi3k* genes remain chemotactic, although they are less polar and do move more slowly in response to cAMP (30, 48). It would appear that PI3K is an important determinant for the establishment of polarity, which streamlines the chemotaxis process, but is not essential. In contrast, mutants that have disruptions in the genes encoding the enzymes responsible for cGMP production, *gca* and *sgc*, are only slightly chemotactic to cAMP (37). Taken together, these results indicate that the generation of cGMP plays a more important role in the chemotactic process than the production of inositol 1,4,5-trisphosphate. A comparison of the rates of chemotaxis of the various *ras* null strains relative to levels of PKB phosphorylation (as a measure of PI3K activity) and cGMP production is consistent with this interpretation. Thus, both chemotaxis and cGMP production were only reduced by a factor of two in *rasG⁻* cells, whereas PKB activation was barely detectable in these cells. Furthermore, the rate of chemotaxis was very similar in *rasC⁻* cells to that of the parental cell line, and cGMP production was only slightly reduced, whereas PKB phosphorylation was markedly reduced. In addition, an examination of the *rasG⁻* cell morphology during chemotaxis revealed little polarity, even though these cells perform relatively efficient chemotaxis (10), consistent with the idea that cGMP pathway is more important for chemotaxis than the PI3K pathway.

There is strong evidence for specificity among the RasGEF proteins, that act upstream of Ras proteins in the cAMP signaling pathway (49). RasGEFA is specific for RasC and appears to be the only RasGEF capable of converting RasC to the GTP-bound form (Fig. 7). RasGEFR is specific for RasG, but because RasG is still partially activated in a *gefR⁻* strain, other RasGEFs (whose identities are currently unknown and, therefore, denoted as *RasGEF-?* in Fig. 7) can also activate RasG. Thus, the input to RasG is more complex than is the input to RasC, which may reflect the fact that RasG has multiple functions in the cell. A key question, still to be examined, is how the activation of the RasGEFs is linked to components of the cAMP receptor complex.

Downstream of RasG and RasC there is clearly a bifurcation of the signaling pathways (Fig. 7). The differential impact of RasG and RasC on ACA activation *versus* chemotaxis is almost certainly due to differences in the interaction of the Ras proteins with their downstream effectors. There is already evidence for differential affinity, because the RBDs of mammalian Raf1 and RalGDS and of *Dictyostelium* PI3K1 and PI3K2 bind more effectively to RasG than to RasC (15). The higher affinity of PI3K1 and PI3K2 for RasG is consistent for the proposed roles for PI3K and RasG in establishing polarity. A yeast two-hybrid analysis has shown that RIP3 also interacts strongly with RasG but does not interact with RasC (50), but the function of RIP3 remains an enigma, because there is a marked discrepancy between this biochemical and the available genetic data. The *rip3*[−] strain is defective in aggregation (50, 51), a phenotype that closely resembles that of the *rasC*[−] and *gefA*[−] strains, and very different from the delayed aggregation phenotype of *rasG*[−] cells. This genetic data, therefore, indicates a close connection between RIP3, RasGEFA, and RasC and provides no evidence for a connection between RIP3 and RasG. The failure of RIP3 to bind to RasC in yeast two-hybrid assays (50) is not simply an artifact, resulting from the failure of RasC to express properly in yeast, because RIP3 binds to RasG but does not bind to RasC in a RBD-RIP3 pulldown assay.⁴ A further complication is that RIP3 is a component of the TORC2 complex, which is necessary for optimum ACA activation and optimum chemotaxis. Understanding these signaling pathways will require a greater understanding of the role(s) of the TORC2 complex. In addition to these affinity considerations, a spatial separation of RasG and RasC may play an important role in the signal bifurcation, because ACA activation occurs at the trailing end of a migrating cell while PI3K stimulation of PKB occurs at the leading end. The study of this activation in live cells is currently hampered by the lack of specific reagents that can uniquely identify activated RasC and activated RasG. Creating such specific reagents will be an important step for future understanding.

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